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M.Sc. QUANTITATIVE FINANCE
SEMESTER III
LECTURE NOTES ON
Machine Learning in Quantitative Finance

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Chapter 1

Machine Learning Foundations & Data Preprocessing

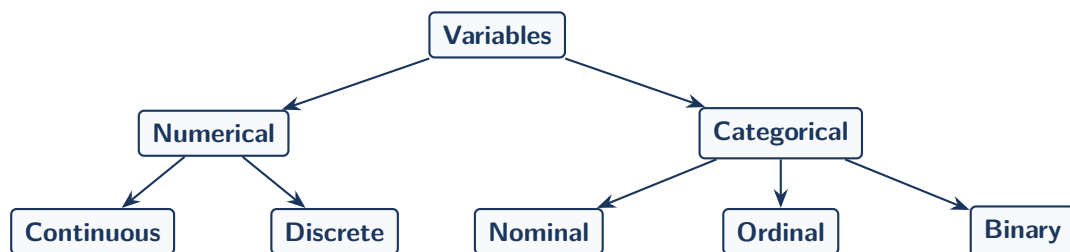
1.1 Challenges in Machine Learning

Machine learning applications in financial engineering face several structural and regulatory constraints:

- **Requires Large, Clean Data:** Models demand substantial volumes of high-quality, cleansed historical records to generalize effectively.
- **Risk of Overfitting:** Complex models risk fitting noise and idiosyncratic market movements rather than genuine underlying signals.
- **Interpretability (“Black Box”):** Advanced algorithms often lack transparent mechanistic explanations, creating auditing hurdles.
- **Regulatory & Ethical Concerns in Finance:** Strict regulatory compliance mandates model explainability, fairness, and accountability in financial decision-making.

1.2 Types of Variables

Variables encountered in machine learning datasets are broadly categorized into numerical and categorical types:

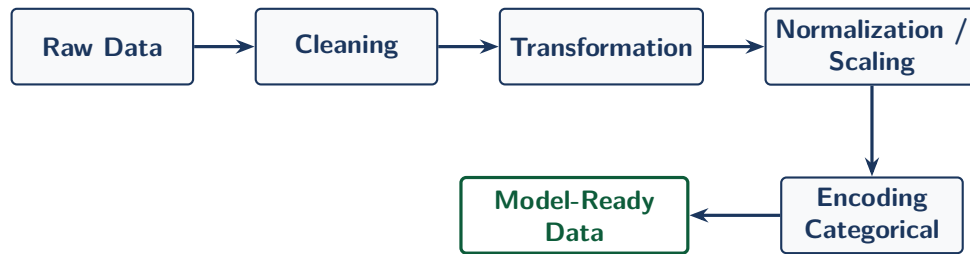


1.3 Data Preprocessing Pipeline

The standard end-to-end data preparation workflow transforms raw financial inputs into model-ready datasets:

Table 1.1: Taxonomy of Data Types and Financial Examples

Type	Subtype	Meaning	Finance Example
Numerical	Continuous	Any value in a continuous range	Stock price, portfolio return, volatility
Numerical	Discrete	Countable whole numbers	Number of trades per day, number of defaults
Categorical	Nominal	No natural ordering among categories	Sector: Banking, IT, Pharma
Categorical	Ordinal	Natural hierarchical order exists	Credit Rating: AAA > AA > A > BBB
Categorical	Binary	Exactly two distinct categories	Loan Default: Yes / No



- **Cleaning:** Removes duplicate records, corrects erroneous entries, and handles missing observations.
- **Transformation:** Converts raw figures into mathematically suitable modeling formats (e.g., calculating log returns from sequential stock prices).
- **Normalization / Scaling:** Brings numerical features across heterogeneous magnitudes into a common comparable scale.
- **Encoding:** Converts non-numeric categorical attributes into numerical indicator representations.

1.3.1 Scaling & Normalization

Features measured in disparate units must be rescaled to ensure distance-based and gradient-based algorithms treat features proportionately:

Min-Max Scaling

Rescales feature values into the bounded closed interval $[0, 1]$:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \quad (1.1)$$

Z-Score Standardization

Standardizes observations using empirical sample mean and sample standard deviation:

$$Z = \frac{x - \mu}{\sigma} \quad (1.2)$$

1.4 Train-Test-Validation Splitting

Datasets are partitioned into distinct subsets to evaluate model performance on unseen data, verifying whether the algorithm generalizes rather than memorizing training samples:

- **Training Set:** Used to fit model parameters and learn patterns.
- **Validation Set:** Used during model development to tune hyperparameters and guide model selection.
- **Testing Set:** Used strictly for final unbiased model performance evaluation.

1.4.1 Common Partitioning Ratios

- **Train-Test Splits:** 80% – 20% or 70% – 30%.
- **Train-Validation-Test Splits:** 70% – 15% – 15%.

1.4.2 K-Fold Cross-Validation

In k -fold cross-validation:

1. The training dataset is partitioned into k equal parts (“folds”).
2. The model is trained on $k - 1$ folds and validated on the remaining fold.
3. This procedure is repeated k times, rotating the held-out validation fold sequentially across all splits.

Benefits & Common Choices

- Provides a significantly more robust estimate of true out-of-sample model performance.
- Reduces vulnerability to an anomalous single lucky or unlucky split.
- Common configurations: $k = 5$ or $k = 10$.

1.4.3 Data Leakage Prevention

Data Leakage Warning

Never scale or normalize the entire dataset prior to splitting.

If sample mean μ and standard deviation σ are calculated over the aggregate dataset, information from the test set leaks directly into the training stage.

Mandatory Pipeline Sequence:

1. Split the raw dataset first.
2. Fit the scaler/imputer exclusively on training data.
3. Transform validation and test data using the saved training parameters.

1.4.4 Splitting Financial Time-Series Data

For time-series data like stock prices, a purely random split is fundamentally problematic because it mixes past and future observations. Sequential financial records must strictly use **chronological splits**.

1.5 Missing Data Mechanisms & Imputation

1.5.1 Missingness Mechanisms

Table 1.2: Mechanisms of Missing Data and Practical Financial Examples

Type	Meaning	Finance Example
MCAR	Missing Completely at Random: Missingness is entirely independent of observed and unobserved variables.	Trade entry terminal intermittently experiences random network packet drops logging trades.
MAR	Missing at Random: Missingness is statistically related to other observed features in the dataset.	Younger retail banking customers are systematically more likely to leave the optional income field blank.
MNAR	Missing Not at Random: Missingness is directly correlated with the actual missing value itself.	Highly indebted distressed borrowers deliberately omit outstanding secondary credit obligations.

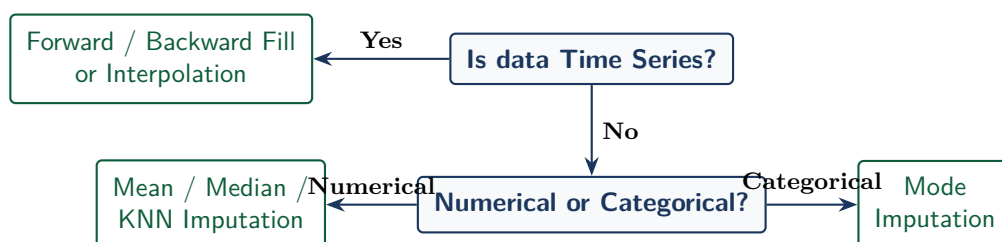
1.5.2 Imputation Methods

Common techniques for replacing missing data include: • Mean / Median / Mode, • K-Nearest Neighbors (KNN) Imputation, • Forward Fill, • Backward Fill, and • Interpolation.

Table 1.3: Advantages and Disadvantages of Imputation Methods

Method	Advantages	Disadvantages
Mean / Median	Simple, fast to compute	Distorts sample variance; ignores inter-feature relationships.
Mode	Directly applicable to categorical data	Can artificially over-represent the most common category.
KNN	Exploits multivariate relationships between features	Computationally expensive on large financial datasets.
Forward / Backward Fill	Well-suited for time-series sequences	Assumes most recent observed value remains representative.
Interpolation	Produces smooth estimates capturing trends	Unsuitable for non-sequential, unordered data.

1.5.3 Decision Framework for Selecting Imputation Methods



1.6 Imbalanced Categorical Data

Class imbalance arises when one category of the target variable appears far more frequently than other categories:

Financial Example

In credit card fraud detection:

- Genuine Transactions: 99.9%
- Fraudulent Transactions: 0.1%

A naive model that blindly predicts “no fraud” every single time achieves 99.9% raw accuracy, yet is entirely useless. Thus, **accuracy alone is fundamentally misleading**.

1.6.1 Consequences of Class Imbalance

- Biases model parameters heavily toward majority class patterns.
- Renders overall accuracy metric uninformative for decision-makers.

1.6.2 Remedial Solutions

- **Oversampling**: Duplicating or synthetically generating minority class instances.
- **Undersampling**: Down-sampling majority class observations.
- **SMOTE**: Synthetic Minority Over-sampling Technique, interpolating new instances along minority feature lines.

1.7 Model Evaluation Metrics

For binary classification problems, predictions and actual outcomes are summarized in a confusion matrix:

Table 1.4: Binary Classification Confusion Matrix

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

Precision

Of all instances predicted as positive, how many were genuinely correct?

$$\text{Precision} = \frac{TP}{TP + FP} \quad (1.3)$$

Recall

Of all actual positive instances in the population, how many did the model identify?

$$\text{Recall} = \frac{TP}{TP + FN} \quad (1.4)$$

F_1 -Score

Harmonic mean of Precision and Recall, balancing false alarms against missed detections:

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (1.5)$$

1.8 Mistakes to be Avoided in ML Pipelines

1. **Global Scaling Before Splitting:** Scaling or normalizing the entire dataset prior to train/test partitioning causes data leakage.
2. **Blind Mean Imputation on MNAR Data:** Imputing missing values with unconditional means when data is Missing Not at Random injects substantial estimation bias.
3. **Ignoring Class Imbalance:** Relying exclusively on overall classification accuracy in skewed datasets conceals poor minority detection.
4. **Unchecked One-Hot Encoding on High-Cardinality Variables:** Creating dummy variables for high-cardinality categorical features leads to severe dimensionality inflation.
5. **Failing to Shuffle Non-Time-Series Data:** Neglecting random shuffling before partitioning standard cross-sectional data risks clustering-induced bias.

1.9 Key Takeaways

Core Principles of ML Data Preparation

- i) **Pre-processing dominates:** Data preprocessing accounts for 60% – 80% of total machine learning workflow effort.
- ii) **Always split first:** Partition data into training, validation, and test sets before performing any scaling or modeling.
- iii) **Diagnose missingness:** Determine whether missing values are MCAR, MAR, or MNAR before selecting an imputation strategy.
- iv) **Address class imbalance:** Use SMOTE, oversampling, or undersampling; accuracy alone is never sufficient.
- v) **Respect temporal order:** For financial and time-series data, strictly employ chronological splits and forward-looking imputation to prevent look-ahead bias.